

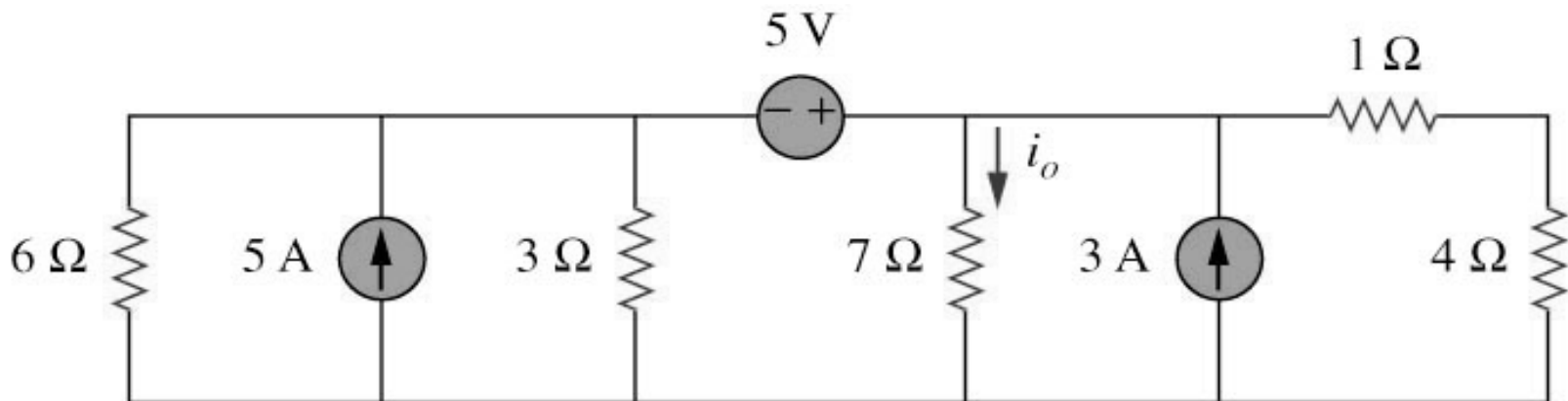
Superposition and Source Transformation

Video 6 Electrical Science

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Superposition and Source Transformation

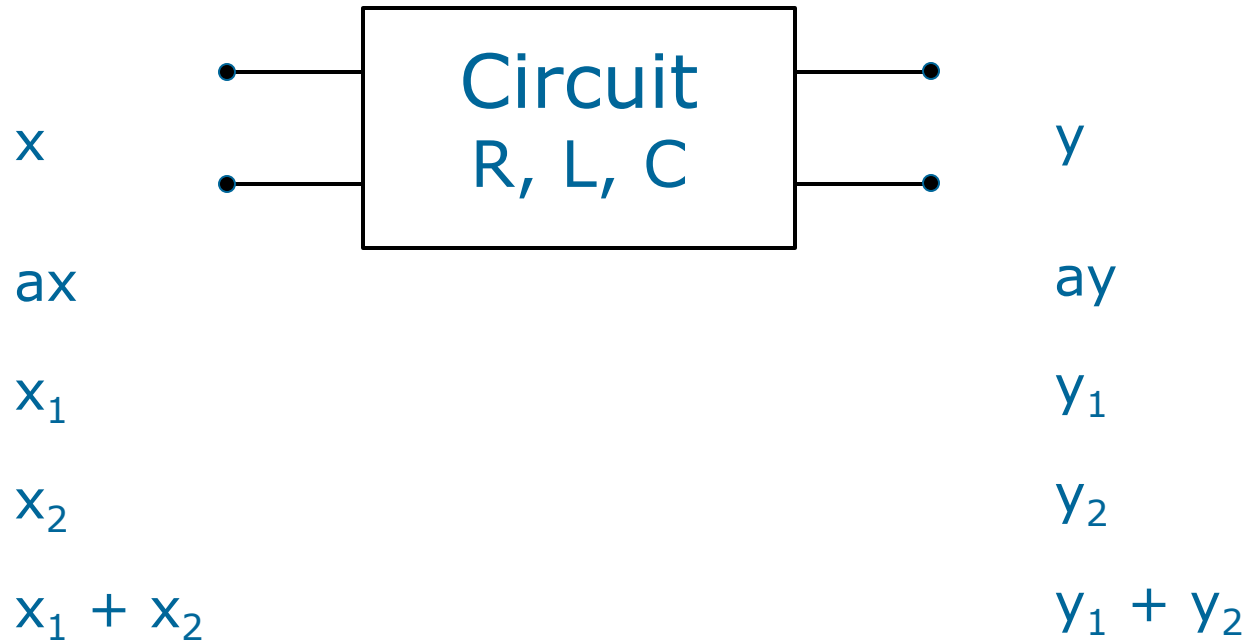
- Linearity Property
- Superposition Theorem
- Source Transformation



Circuit Theory Video 6

Linearity Property

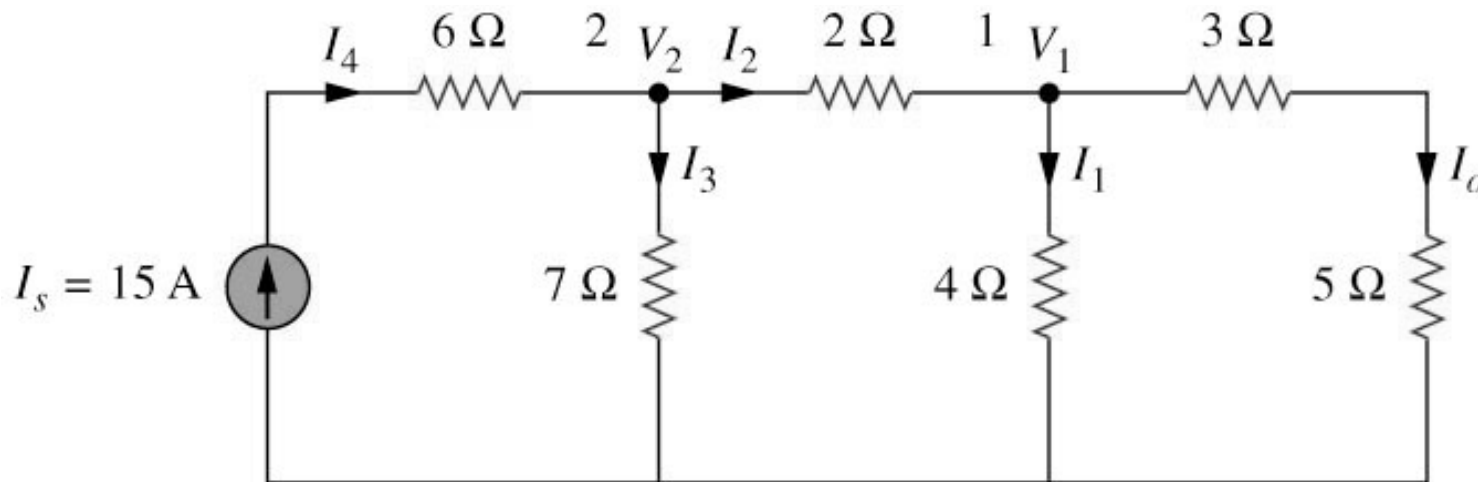
A linear circuit is one whose output is linearly related (or directly proportional) to its input.



Linearity Property

Example 1

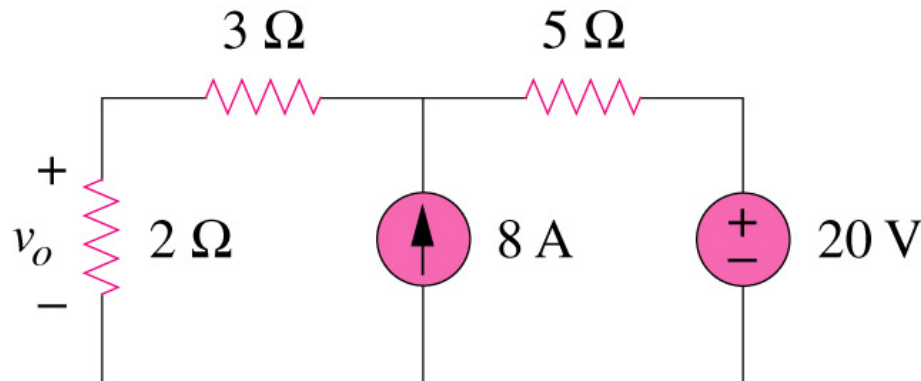
By assume $I_o = 1$ A, use linearity to find the actual value of I_o in the circuit shown below.



Answer $I_o = 3$ A

Superposition Theorem

It states that the voltage across (or current through) an element in a linear circuit is the algebraic sum of the voltage across (or currents through) that element due to EACH independent source acting alone.

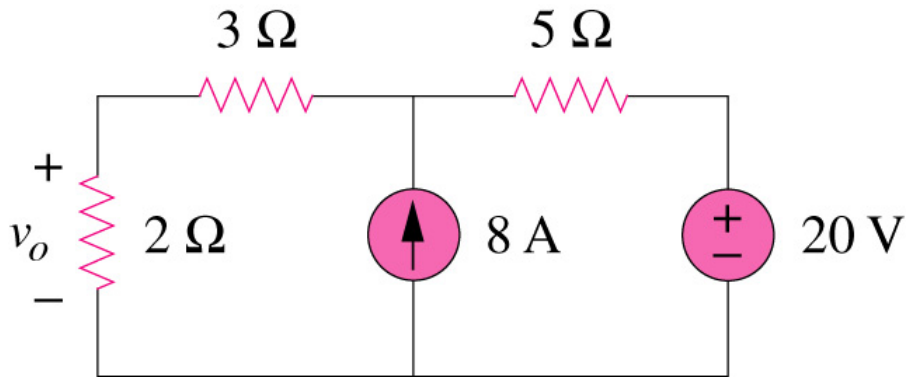


- Independent voltage sources are replaced by 0 V (short circuit)
- Independent current sources are replaced by 0 A (open circuit).
- Dependent sources are left intact because they are controlled by circuit variables.

Superposition Theorem

Example 2:

Use the superposition theorem to find v_o in the circuit.

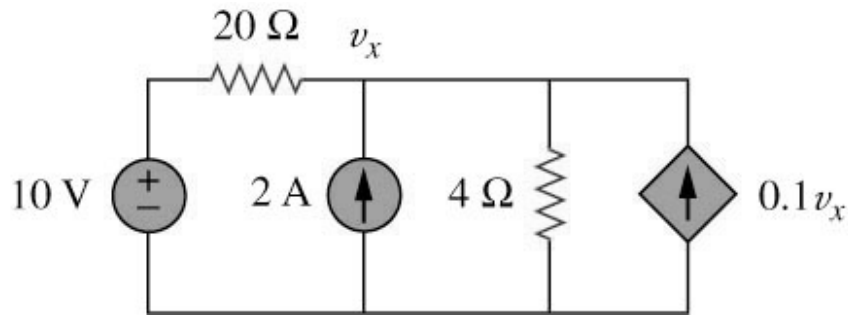


Answer $V_o = 12\text{V}$

Superposition Theorem

Example 3

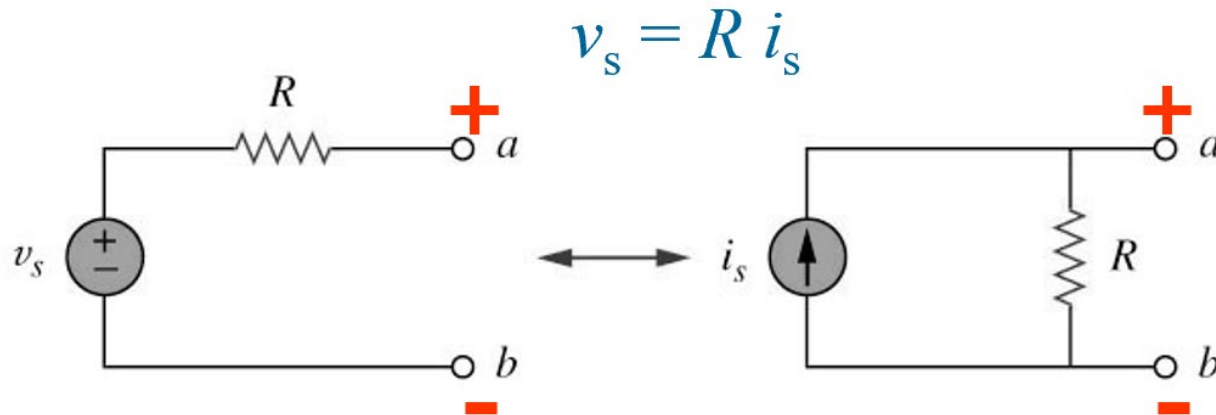
Use superposition to find v_x in the circuit below.



Answer $V_x = 12.5\text{V}$

Source Transformation

It is the process of replacing a voltage source v_s in series with a resistor R by a current source i_s in parallel with a resistor R , or vice versa.

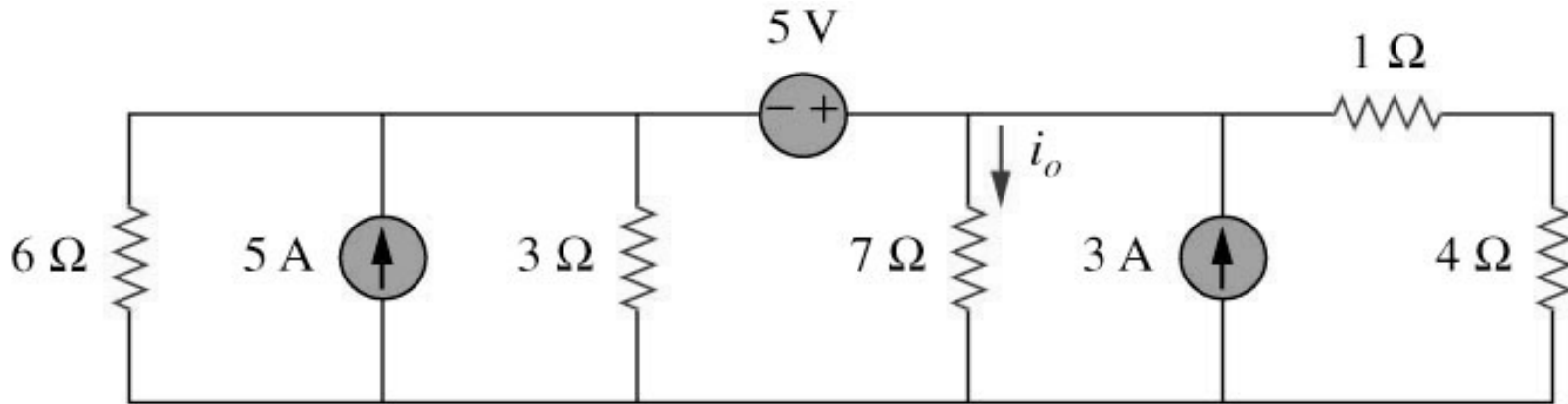


- The arrow of the current source is directed toward the positive terminal of the voltage source.
- Source transformation apply to dependent sources too

Source Transformation

Example 4

Find i_o in the circuit shown below using source transformation.



Answer $i_o = 1.78\text{ A}$